

Topic 2: A Brief History of AI

2.1 Early AI (1950s--1980s)

- 1950: Alan Turing proposes the Turing Test -- can a machine exhibit intelligent behaviour indistinguishable from a human?
- 1956: the term 'Artificial Intelligence' is coined at the Dartmouth Conference by John McCarthy and colleagues
- 1960s--70s: rule-based expert systems -- AI encoded as hand-written IF-THEN rules
- First AI Winter (1974--1980): overhyped promises went unmet; research funding dried up
- 1980s: expert systems revival -- useful in narrow domains but too brittle and expensive to scale
- Second AI Winter (1987--1993): expert systems failed to generalise; investment collapsed again

2.2 Machine Learning Era (1990s--2012)

- Shift from hand-coded rules to learning patterns from data automatically
- 1997: IBM Deep Blue defeats world chess champion Garry Kasparov
- 2006: Geoffrey Hinton revives deep neural networks with deep belief networks
- 2009: ImageNet dataset published -- 1.2 million labelled images enabling computer vision benchmarks
- 2012: AlexNet wins the ImageNet competition by a large margin using deep learning, launching the deep learning era
- 2014: Generative Adversarial Networks (GANs) introduced by Ian Goodfellow -- generators and discriminators compete to produce realistic images

2.3 The Road to Modern Generative AI (2017--2022)

- 2017: 'Attention is All You Need' paper by Vaswani et al. introduces the Transformer architecture (arxiv.org/abs/1706.03762)
- 2018: BERT (Google) and GPT-1 (OpenAI) -- large language models pre-trained on text and fine-tuned for tasks
- 2019: GPT-2 (1.5 billion parameters) -- so capable OpenAI initially staged its release over safety concerns
- 2020: GPT-3 (175 billion parameters) -- demonstrated in-context learning and remarkably general capabilities
- 2021: DALL-E and Codex released; AlphaFold 2 (DeepMind) predicts protein structures with high accuracy
- 2022: Stable Diffusion (open source), ChatGPT, and Midjourney bring generative AI to a mainstream audience

2.4 The Current Era (2023--Present)

- 2023: GPT-4 (multimodal), Claude 2 (Anthropic), Gemini (Google) -- near-human performance on many professional benchmarks
 - 2023: Llama 2 (Meta) released as open weights -- enabling the open-source AI ecosystem to grow rapidly
 - 2024: smaller but highly capable models (Llama 3, Mistral, Phi-3) that run on consumer hardware
 - 2024--2025: AI agents -- models that plan, use external tools, browse the web, and complete multi-step tasks
 - The pace of capability improvement is accelerating; abilities that seemed years away have arrived in months
-